

Graham Anderson

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U.S. citizen. Extensive experience delivering under export-controlled (ITAR) constraints.

HEADLINE

Principal AI Engineer | Machine Learning Engineer | AI Architect | Agentic AI · LLM/RAG · Knowledge Graphs | Defense & Aerospace | Founder, grahamaco | Buffalo, NY · Remote

ABOUT

I build verifiable agentic AI systems for defense, aerospace, and compliance. On DARPA ARCOS, I was Principal Data Scientist and ACERT Technical Lead for CS Group's contribution: the responsible technical/project lead for knowledge-graph and LLM reasoning delivery for automated certification of mission-critical software, working alongside Honeywell, Lockheed Martin, MIT, GE Research, SRI, and other program collaborators.

Much of my current client work is export-controlled, so client and program details are withheld. Publicly releasable independent engineering is at github.com/grahama1970:

- tau — receipt-gated multi-agent harness: agent work compiles to typed DAG contracts; every handoff needs a schema-valid receipt or validator result before the next step runs. No receipt, no action.
- pdf_oxide — my heavily diverged fork of yfedoseev/pdf_oxide (independent since Mar 2026): 430 commits and ~137K lines added, including Rust-core changes, a full Python pipeline/plugin system, PDF-cloning fixture generation, extraction calibration, and NIST document-validation tooling.
- agent-skills — 330+ reusable agent skills and 70+ bounded worker roles, portable across Claude Code, Codex, and Gemini. Public repo, private runtime — effectively my living resume.
- Private, regulated side: an ArangoDB agent-memory platform — approximately 219K evidence-grounded question–reasoning–answer records across 7K+ security-control records (NIST 800-53/171, CWE, MITRE ATT&CK, D3FEND, SPARTA), hybrid BM25 + vector + graph recall over 2.2M source chunks; a corpus of ~94K Lean 4 theorem statements indexed for requirements-to-proof retrieval.

Recent public work includes a receipt-gated multi-agent harness, a Rust PDF extraction platform, and a large library of bounded agent capabilities. Presented Agentic Cybersecurity research to Air Force Research Laboratory and received an AFRL "Hacker" challenge coin following the presentation.

15+ years of hand-coding underneath it all — today I work primarily through agentic coding, driving many harnesses and models, including my own (tau). I build the tools that make agentic development trustworthy, and I use them daily.

From simulated worlds to mission-critical verifiable AI: Executive Producer on the God of War: Ascension interactive campaign (Real Pie, for Sony; Webby-recognized), leading 80+ person interactive productions; before that, commercial composer for major consumer brands (Adidas, Pepsi, X-Games). Creative systems thinking is why my AI systems are architected, not stapled together.

Buffalo, NY based (hybrid/onsite there, or remote). Open to contract engagements and full-time Principal/Staff AI Engineer, AI Architect, Machine Learning Engineer, LLM Platform, and Security/Compliance AI roles.

EXPERIENCE

Founder & Principal AI Engineer / Architect | grahamaco | Buffalo–Niagara Falls Area · Remote

Feb 2025 - Present Independent AI engineering practice. Clients are primarily export-controlled (ITAR) defense/aerospace — names withheld; publicly releasable engineering is at github.com/grahama1970.

- Built tau, a receipt-gated multi-agent orchestration harness: goals compile to typed DAG contracts; every agent handoff must produce a schema-valid receipt, hashed evidence artifact, or validator result — no receipt, no action.
- Develop a heavily diverged fork of pdf_oxide (origin: yfedoseev/pdf_oxide, MIT/Apache-2.0; independent since Mar 2026): 430 commits, ~137K lines added — Rust-core changes, full Python pipeline/plugin system, PDF-cloning fixture generation, extraction calibration, and NIST document-validation tooling.
- Authored agent-skills: 330+ reusable agent capabilities (compliance evidence mapping, document AI, LLM evaluation, retrieval) and 70+ bounded worker roles — public repo, private runtime.
- Built an ArangoDB agent-memory platform (private, regulated): ~219K evidence-grounded QRA records across 7K+ security-control records; hybrid BM25 + vector + multi-hop graph recall over 2.2M source chunks; ~94K Lean 4 theorem statements indexed for requirements-to-proof retrieval.
- Core contributor to pi-mono (open-source agent framework/TUI): extension system, agent runtime, and TUI harness work, Feb–Jul 2026.

Lead Research Scientist (Agentic Formal Methods) | Self-Employed / Various Clients | Buffalo, NY

Jan 2024 - Feb 2025

- Designed verifiable agentic systems: a process-driven autoformalization workflow translating engineering requirements into Lean 4 proofs for safety-critical workflows.
- Built probabilistic-deterministic self-correcting loops where LLM generation is validated by compiler/prover feedback.
- Consulted in aerospace/defense settings under export-controlled / ITAR constraints.
- Presented Agentic Cybersecurity work to Air Force Research Laboratory; received an AFRL "Hacker" challenge coin.

Principal Data Scientist & ACERT Technical Lead | CS Group (DARPA ARCOS)

Sep 2020 - Dec 2023

- Responsible technical/project lead for CS Group's contribution to the 4-year DARPA ARCOS program (Automated Rapid Certification of Software), working alongside Honeywell, Lockheed Martin, MIT, GE Research, SRI, and other program collaborators.
- Led design and delivery of ACERT (Automated Certification of Requirements Tool): architected the ArangoDB knowledge-graph schema and LLM-assisted reasoning pipeline for multi-hop compliance verification of mission-critical software against complex certification standards.
- Worked with government and aerospace program stakeholders; split time roughly 50/50 between executive/technical briefings and hands-on architecture and code.

Executive Producer / Marketing Executive / Professional Composer | Los Angeles, CA

Mar 2005 - 2016

- Executive Producer, God of War: Ascension interactive campaign (Real Pie, for Sony; Webby-recognized); led cross-functional teams (80+) on large-scale interactive productions.
- Commercial composer for major consumer brands (Adidas, X-Games, Pepsi, Taco Bell, Nestle).

PUBLIC WORK (non-ITAR) — github.com/grahama1970

Client work is mostly export-controlled, so here is the public, verifiable side — including the fun stuff:

- agent-skills — my living resume: 330+ reusable agent skills, 70+ worker roles. Public repo, private runtime. <https://github.com/grahama1970/agent-skills>

- tau — receipt-gated multi-agent harness. "Agents hallucinate. Tau contains them." <https://github.com/grahama1970/tau>
- pdf_oxide — heavily diverged fork of yfedoseev's Rust PDF toolkit (430 commits, ~137K lines added: Rust-core changes, Python pipeline, PDF cloning, NIST validation). https://github.com/grahama1970/pdf_oxide
- scillm, extractor, anvil, fetcher, chatterbox voice-agent fork — supporting cast, all public.

TOP SKILLS

Machine Learning, Generative AI, Large Language Models (LLM), Agentic AI, AI Agents, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Knowledge Graphs, GraphRAG, ArangoDB, Vector Databases, Prompt Engineering, LLM Evaluation, LLMOps, Formal Verification, Lean 4, Python, Rust, Model Context Protocol (MCP), NIST Frameworks, AI Governance, Docker, Linux